

Clinical Reference Guide: Colorectal Cancer

Extracted Focus & Question-Answer Summary from Cancer.gov PDQ

DOCUMENT OVERVIEW

Focus Condition:	Colorectal Cancer
Semantic Group:	Disorders (T191)
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Question 1: What is (are) Colorectal Cancer ?

Colorectal cancer is a disease in which malignant (cancer) cells form in the tissues of the colon or the rectum. Both organs are vital parts of the body's digestive system, which processes nutrients (vitamins, minerals, carbohydrates, fats, proteins, and water) from foods and helps pass waste material out of the body. The digestive system comprises the mouth, throat, esophagus, stomach, and small and large intestines. The colon (large bowel) represents the first section of the large intestine and is approximately 5 feet long. The rectum and anal canal make up the final 6 to 8 inches of the large intestine, terminating at the anus.

Key Clinical Facts:

- **Terminology:** Cancer starting in the colon is called colon cancer, while cancer originating in the rectum is called rectal cancer. Malignancies affecting either organ are collectively referred to as colorectal cancer.
- **Mortality & Demographics:** Colorectal cancer is the second leading cause of cancer deaths in the United States and is diagnosed more frequently in men than in women. Furthermore, the number of new cases and deaths is higher in African Americans compared to other races.
- **Epidemiological Trends:** The overall number of new colorectal cancer cases and deaths has been decreasing slightly each year. However, in adults younger than 50 years old, there has been a steady, slow annual increase in new cases since 1998.
- **Early Detection:** Finding and treating colorectal cancer early through screening tests can prevent disease-related mortality.

Question 2: Who is at risk for Colorectal Cancer? ?

An individual's likelihood of developing colorectal cancer is governed by a balance of established risk factors and protective factors:

Established Risk Factors (Increase Risk)

- **Age:** Risk increases significantly after age 50, when the majority of cases are diagnosed.
- **Family History:** Having a parent, sibling, or child with colorectal cancer doubles a person's risk.

- **Personal Medical History:** Risk is elevated in individuals with previous colorectal cancer, high-risk adenomas (polyps ≥ 1 cm or exhibiting abnormal histology), ovarian cancer, or chronic inflammatory bowel disease (such as ulcerative colitis or Crohn's disease).
- **Inherited Genetic Risk:** Carrying specific gene mutations linked to familial adenomatous polyposis (FAP) or hereditary nonpolyposis colon cancer (HNPCC / Lynch Syndrome) markedly increases risk.
- **Alcohol Consumption:** Drinking 3 or more alcoholic beverages daily increases colorectal cancer risk and is linked to the formation of large benign colorectal adenomas.
- **Cigarette Smoking:** Associated with an increased risk of developing adenomas, colorectal cancer, and disease-specific death. Smokers who undergo adenoma removal face a higher risk of polyp recurrence.
- **Obesity:** Strongly linked to both increased incidence of colorectal cancer and higher mortality rates.

Established Protective Factors (Decrease Risk)

- **Physical Activity:** Maintaining a lifestyle that includes regular physical activity reduces colorectal cancer risk.
- **Aspirin Use:** Studies indicate that regular aspirin intake lowers both the incidence and mortality of colorectal cancer, with risk reductions typically emerging 10 to 20 years after initiation. However, daily or alternate-day use (≤ 100 mg) carries potential harms, including gastrointestinal bleeding and hemorrhagic stroke, particularly among elderly patients or those with bleeding diatheses.
- **Combination Hormone Replacement Therapy (HRT):** Estrogen-progestin HRT lowers invasive colorectal cancer risk in postmenopausal women. However, if cancer does develop, it is more likely to be diagnosed at an advanced stage, and overall mortality is not reduced. Combination HRT also carries serious risks, including breast cancer, heart disease, and blood clots.
- **Polyp Removal:** Because most polyps are adenomas that can transform into malignancy, endoscopic removal of adenomas larger than 1 cm (pea-sized) lowers colorectal cancer risk. Potential complications of polypectomy include colonic perforation and bleeding.

Factors With Unclear or No Effect

- **Unclear Effect:** It remains unproven whether diet (low-fat/high-fiber vs. high-fat/meat), calcium supplements, or non-aspirin NSAIDs (e.g., sulindac, celecoxib, naproxen, ibuprofen) directly reduce colorectal cancer risk. While celecoxib and sulindac reduce polyp recurrence and polyp burden in FAP patients, their definitive impact on invasive cancer risk is not yet established, and NSAIDs carry renal, cardiovascular, and gastrointestinal bleeding risks.
- **No Effect:** Neither estrogen-only HRT nor statins (cholesterol-lowering drugs) affect the risk of developing or dying from colorectal cancer.

Question 3: How to prevent Colorectal Cancer ?

Primary cancer prevention relies on actively avoiding known modifiable risk factors while enhancing established protective factors.

Patients can proactively reduce their risk by quitting cigarette smoking, maintaining a healthy body weight through balanced nutrition and regular physical exercise, moderating alcohol intake, and undergoing appropriate endoscopic screening for polyp removal. Individuals should consult their healthcare providers to discuss tailored preventive strategies, including the risk-benefit profiles of chemopreventive agents such as aspirin.

Question 4: What research (or clinical trials) is being done for Colorectal Cancer ?

Cancer prevention clinical trials are actively conducted to study scientific methods for reducing the risk of developing colorectal cancer. These studies investigate whether specific lifestyle interventions, medications, vitamins, minerals, or dietary supplements can effectively prevent malignancy.

Prevention trials enroll diverse participant groups, including healthy individuals without known risk factors, healthy people with elevated risk profiles, and cancer survivors aiming to prevent secondary tumors or recurrences. Ongoing clinical trials across the country are continually evaluating novel preventive approaches for both colon and rectal cancers.

This document was generated for educational and clinical reference purposes based on official patient treatment guidelines (PDQ) from the National Cancer Institute.